Agentic Systems

Institute: Indian Institute of Information Technology (IIIT) Surat
Participants: Senior Undergrads.
Lab: deepdynamicsai.github.io **Coordinator:** Dr. Pradeep Singh
Format: Weekly reading, presentation, discussion, implementation, and research meetings

Objective

The project is designed as a structured transition from *understanding agentic systems* to *formulating and solving a research problem.* The first six months will therefore emphasize conceptual depth rather than rapid framework-based development. Students should understand an agent as a system combining

$$\boxed{\text{model} + \text{state/context} + \text{tools} + \text{control policy} + \text{environment} + \text{feedback}}$$

and should be able to explain not only how an agent works, but also why it fails.

Progression

Months 1–2: Foundations and Agent Loops. Agents and environments; LLMs as controllers; structured outputs and tool calls; RAG versus memory versus tools; deterministic workflows versus agents; ReAct; construction of a minimal agent loop in plain Python without relying initially on agent frameworks.

Months 3–4: Planning, Reflection and Memory. Task decomposition; planning and replanning; search over reasoning trajectories; reflection and self-correction; episodic, semantic, procedural and working memory; retrieval, forgetting, compression and context engineering; agentic RAG and reusable skills.

Months 5–6: Multi-Agent Systems, Protocols and Evaluation. Delegation, specialization and coordination; multi-agent communication; human-in-the-loop control; stateful agent graphs; MCP and agent interoperability; evaluation of trajectories rather than only final answers; reliability, observability, prompt injection, permissions and agent security. Frameworks such as LangGraph, AutoGen or similar systems may be studied only after the underlying mechanisms are understood.

Month 7: Research Problem Selection. The group will reproduce small baselines, deliberately generate failures, build a failure taxonomy, and study the nearest literature. One of the following directions will then be selected:

- **Adaptive Deliberation and Failure-Aware Control.** Can an agent decide *when* to continue reasoning, re-plan, call another tool, verify an intermediate result, or stop? The aim is to design a lightweight controller that allocates additional computation only when the trajectory indicates likely failure, and to study the resulting accuracy–cost–latency trade-off.
- **Reliable Long-Term Memory under a Fixed Context Budget.** What should an agent remember, compress, update, or deliberately forget? The project will study memory utility, stale or contradictory memories, retrieval errors and memory contamination, with the goal of developing a principled memory-selection/forgetting mechanism for persistent agents.

Month 8: Research Execution. The selected problem will be converted into a precise hypothesis, baselines, proposed method, evaluation protocol and ablation study. The final deliverables will include reproducible code, failure analysis, experimental results, a technical report, and a research presentation.

Weekly Working Method and Reading Spine

Each meeting will contain a concept presentation, one primary paper, a minimal implementation or worked example, and a discussion devoted specifically to assumptions and failure modes. Presenters will be paired with a *challenger* whose role is to identify weaknesses, counterexamples and unjustified claims. A common repository will maintain slides, two-page notes, code and a continuing "failure book."

Core readings will include surveys of LLM-based autonomous agents; and accessible primary papers such as *ReAct*, *Toolformer*, *Reflexion*, *Generative Agents*, planning and memory surveys, followed by papers on multi-agent systems, evaluation and security.

Guiding principle: *Understand → Implement → Break → Measure → Explain → Research.* The objective is not merely to assemble an agent using existing frameworks, but to identify and study a reproducible limitation of contemporary agentic systems.